

PROJECT FILE

WORKING WITH MONGO DB DOCUMENTS AND COLLECTIONS

SUBMITTED TO – _MR HARENDRA SINGH_ RAJPOOT

ROLL NO -1250258268 (26)

SUBMITTED BY – Mohammad Rehan Ansari

BCA – Data Science & Artificial Intelligence

SEMESTER – ODD (3rd)

UNIVERSITY / COLLEGE – BBD UNIVERSITY

CREATE OPERATION- The **Create** operation is used to add new documents to a collection. MongoDB provides methods like `insertOne()` for single documents and `insertMany()` for multiple documents.

Read Operation- The Read operation retrieves documents from a collection. MongoDB uses methods like `find()` to fetch multiple documents and `findOne()` to retrieve a single document.

```
> use("CRUD")
< switched to db CRUD
> db["CRUD"].find()
<
> db.CRUD.insertOne({rollno:1,name:"Sam",age:20,department:ECA})
✖ ReferenceError: ECA is not defined
> db.CRUD.insertOne({rollno:1,name:"Sam",age:20,department:"ECA"})
< {
  acknowledged: true,
  insertedId: ObjectId('6aae544e05178ee015c9caf2')
}
> db.CRUD.find()
< {
  _id: ObjectId('6aae544e05178ee015c9caf2'),
  rollno: 1,
  name: 'Sam',
  age: 20,
  department: 'ECA'
}
> db.CRUD.insertMany([
  {rollno:2,name:"Ram",age:20,department:"ECA"},
  {roll:3,name:"Adi",age:20,department:"ECA"},
  {roll:4,name:"E
```

```
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6aae5e6505178ee015c9caf3'),
    '1': ObjectId('6aae5e6505178ee015c9caf4'),
    '2': ObjectId('6aae5e6505178ee015c9caf5'),
    '3': ObjectId('6aae5e6505178ee015c9caf6'),
    '4': ObjectId('6aae5e6505178ee015c9caf7'),
    '5': ObjectId('6aae5e6505178ee015c9caf8'),
    '6': ObjectId('6aae5e6505178ee015c9caf9'),
    '7': ObjectId('6aae5e6505178ee015c9cafa'),
    '8': ObjectId('6aae5e6505178ee015c9cafb'),
    '9': ObjectId('6aae5e6505178ee015c9caf'),
    '10': ObjectId('6aae5e6505178ee015c9cafd'),
    '11': ObjectId('6aae5e6505178ee015c9cafe'),
    '12': ObjectId('6aae5e6505178ee015c9caff'),
    '13': ObjectId('6aae5e6505178ee015c9cb00'),
    '14': ObjectId('6aae5e6505178ee015c9cb01'),
    '15': ObjectId('6aae5e6505178ee015c9cb02'),
    '16': ObjectId('6aae5e6505178ee015c9cb03'),
    '17': ObjectId('6aae5e6505178ee015c9cb04'),
    '18': ObjectId('6aae5e6505178ee015c9cb05'),
    '19': ObjectId('6aae5e6505178ee015c9cb06'),
    '20': ObjectId('6aae5e6505178ee015c9cb07'),
    '21': ObjectId('6aae5e6505178ee015c9cb08'),
    '22': ObjectId('6aae5e6505178ee015c9cb09'),
    '23': ObjectId('6aae5e6505178ee015c9cb0a'),
    '24': ObjectId('6aae5e6505178ee015c9cb0b'),
    '25': ObjectId('6aae5e6505178ee015c9cb0c'),
    '26': ObjectId('6aae5e6505178ee015c9cb0d'),
    '27': ObjectId('6aae5e6505178ee015c9cb0e'),
    '28': ObjectId('6aae5e6505178ee015c9cb0f'),
    '29': ObjectId('6aae5e6505178ee015c9cb10'),
    '30': ObjectId('6aae5e6505178ee015c9cb11'),
    '31': ObjectId('6aae5e6505178ee015c9cb12'),
    '32': ObjectId('6aae5e6505178ee015c9cb13'),
    '33': ObjectId('6aae5e6505178ee015c9cb14'),
    '34': ObjectId('6aae5e6505178ee015c9cb15'),
    '35': ObjectId('6aae5e6505178ee015c9cb16'),
    '36': ObjectId('6aae5e6505178ee015c9cb17'),
    '37': ObjectId('6aae5e6505178ee015c9cb18'),
    '38': ObjectId('6aae5e6505178ee015c9cb19'),
    '39': ObjectId('6aae5e6505178ee015c9cb1a')
  }
}
```

```

> db.CRUD.find()
< {
  _id: ObjectId('6aae544e05178ee015c9caf2'),
  rollno: 1,
  name: 'Sam',
  age: 20,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6505178ee015c9caf3'),
  rollno: 2,
  name: 'Ram',
  age: 20,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6505178ee015c9caf4'),
  roll: 3,
  name: 'Adi',
  age: 20,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6505178ee015c9caf5'),
  roll: 4,
  name: 'Era',
  age: 20,
  department: 'ME'
}
{
  _id: ObjectId('6aae5e6505178ee015c9caf6'),
  roll: 5,
  name: 'Bee',
  age: 20,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6505178ee015c9caf7'),
  roll: 6,
  name: 'Arif',
  age: 21,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6505178ee015c9caf8'),
  roll: 13,
  name: 'Krish',
  age: 21,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6505178ee015c9caff'),
  roll: 14,
  name: 'Mukesh',
  age: 20,
  department: 'ME'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cb00'),
  roll: 15,
  name: 'Raheem',
  age: 21,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cb01'),
  roll: 16,
  name: 'Kaif',
  age: 22,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cb02'),
  roll: 17,
  name: 'Muskan',
  age: 22,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cb03'),
  roll: 18,
  name: 'Leher',
  age: 21,
  department: 'CSE'
}
}

```

```

{
  _id: ObjectId('6aae5e6505178ee015c9caf8'),
  roll: 7,
  name: 'Adeel',
  age: 21,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6505178ee015c9caf9'),
  roll: 8,
  name: 'Nasib',
  age: 21,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cafa'),
  roll: 9,
  name: 'Arslan',
  age: 20,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cafb'),
  roll: 10,
  name: 'Rehan',
  age: 20,
  department: 'ME'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cafca'),
  roll: 11,
  name: 'Faheen',
  age: 21,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cafcd'),
  roll: 12,
  name: 'Akhil',
  age: 21,
  department: 'ME'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cb04'),
  roll: 19,
  name: 'Ashar',
  age: 21,
  department: 'ME'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cb05'),
  roll: 20,
  name: 'Husnain',
  age: 21,
  department: 'ECA'
}
}
Type "it" for more
it
{
  _id: ObjectId('6aae5e6505178ee015c9cb06'),
  roll: 21,
  name: 'Jhon',
  age: 20,
  department: 'ME'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cb07'),
  roll: 22,
  name: 'Shon',
  age: 22,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cb08'),
  roll: 23,
  name: 'Nikhil',
  age: 22,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cb09'),
  roll: 24,
  name: 'Mitish',
  age: 22,
  department: 'CSE'
}
}

```

```

{
  _id: ObjectId('6aae5e6505178ee015c9cb0a'),
  roll: 25,
  name: 'Mayan',
  age: 20,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cb0b'),
  roll: 26,
  name: 'Piyush',
  age: 21,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cb0c'),
  roll: 27,
  name: 'Ayan',
  age: 21,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cb0d'),
  roll: 28,
  name: 'Faizan',
  age: 21,
  department: 'ME'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cb0e'),
  roll: 29,
  name: 'Palak',
  age: 21,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cb0f'),
  roll: 30,
  name: 'Hazel',
  age: 20,
  department: 'ME'
}
}

```

```

{
  _id: ObjectId('6aae5e6505178ee015c9cb10'),
  roll: 31,
  name: 'Divyanshu',
  age: 21,
  department: 'ME'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cb11'),
  roll: 32,
  name: 'Shivan',
  age: 20,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cb12'),
  roll: 33,
  name: 'Ankush',
  age: 20,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cb13'),
  roll: 34,
  name: 'Anan',
  age: 22,
  department: 'ME'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cb14'),
  roll: 34,
  name: 'Abhinay',
  age: 20,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cb15'),
  roll: 35,
  name: 'Vinay',
  age: 20,
  department: 'CSE'
}
}

```

```

{
  _id: ObjectId('6aae5e6505178ee015c9cb16'),
  roll: 36,
  name: 'Akash',
  age: 21,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cb17'),
  roll: 37,
  name: 'Anmol',
  age: 20,
  department: 'ME'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cb18'),
  roll: 38,
  name: 'Kunal',
  age: 21,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cb19'),
  roll: 39,
  name: 'Anil',
  age: 22,
  department: 'CSE'
}
}
Type "it" for more
it
{
  _id: ObjectId('6aae5e6505178ee015c9cb1a'),
  roll: 40,
  name: 'Nitya',
  age: 21,
  department: 'CSE'
}
}

```

UPDATE OPERATION- The Update operation modifies existing documents.

MongoDB provides methods like update One() to update a single document and update Many() to update multiple documents. The \$set operator is commonly used to specify the fields to update.

```
db.CRUD.updateOne({roll:31},{set:{department:"ECA"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
{
  _id: ObjectId('6aae5e6505178ee015c9cb10'),
  roll: 31,
  name: 'Divyanshu',
  age: 21,
  department: 'ECA'
}
```

```
db.CRUD.updateMany({roll:22},{set:{department:"ME"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
db.CRUD.find()
```

```
{
  _id: ObjectId('6aae5e6505178ee015c9cafb'),
  roll: 10,
  name: 'Rehan',
  age: 20,
  department: 'ME'
}
{
  _id: ObjectId('6aae5e6505178ee015c9caf0'),
  roll: 11,
  name: 'Faheen',
  age: 21,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cafd'),
  roll: 12,
  name: 'Akhil',
  age: 21,
  department: 'ME'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cafe'),
  roll: 13,
  name: 'Krish',
  age: 21,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6505178ee015c9caff'),
  roll: 14,
  name: 'Nukesh',
  age: 20,
  department: 'ME'
}
```

DELETE OPERATION- The Delete operation removes documents from a collection. MongoDB offers deleteOne() to delete a single document and deleteMany() to remove multiple documents based on a filter.

```
> db.CRUD.deleteOne({roll:15})
< {
  acknowledged: true,
  deletedCount: 1
}
> db.CRUD.find()
```

```
db.CRUD.deleteMany({department:"ME"})
{
  acknowledged: true,
  deletedCount: 16
}
db.CRUD.find()
```

```
db.CRUD.find()
{
  _id: ObjectId('6aae544e05178ee015c9caf2'),
  rollno: 1,
  name: 'Sam',
  age: 20,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6505178ee015c9caf3'),
  rollno: 2,
  name: 'Ram',
  age: 20,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6505178ee015c9caf4'),
  roll: 3,
  name: 'Adi',
  age: 20,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6505178ee015c9caf6'),
  roll: 5,
  name: 'Bee',
  age: 20,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6505178ee015c9caf7'),
  roll: 6,
  name: 'Arif',
  age: 21,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6505178ee015c9caf8'),
  roll: 7,
  name: 'Adeel',
  age: 21,
  department: 'CSE'
}
```

```
{
  _id: ObjectId('6aae5e6505178ee015c9caff'),
  roll: 14,
  name: 'Mukesh',
  age: 20,
  department: 'ME'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cb01'),
  roll: 16,
  name: 'Kaif',
  age: 22,
  department: 'ME'
}
```

```
{
  _id: ObjectId('6aae5e6505178ee015c9caf9'),
  roll: 8,
  name: 'Nasib',
  age: 21,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cafa'),
  roll: 9,
  name: 'Arslan',
  age: 20,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cafca'),
  roll: 11,
  name: 'Faheem',
  age: 21,
  department: 'ECA'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cafe'),
  roll: 13,
  name: 'Krish',
  age: 21,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cb03'),
  roll: 18,
  name: 'Leher',
  age: 21,
  department: 'CSE'
}
{
  _id: ObjectId('6aae5e6505178ee015c9cb05'),
  roll: 20,
  name: 'Husnain',
  age: 21,
  department: 'ECA'
}
```

Using greater than in Mongo DB

```
> db.students.find({marks:{$gt:80}})
< {
  _id: ObjectId('6ab21b30555468477e03bd2e'),
  name: 'Amit',
  age: 21,
  marks: 85,
  department: 'CSE',
  skills: [
    'Python',
    'MongoDB',
    'Java'
  ]
}
{
  _id: ObjectId('6ab21b30555468477e03bd30'),
  name: 'Priya',
  age: 20,
  marks: 91,
  department: 'CSE',
  skills: [
    'Java',
    'MongoDB',
    'AI'
  ]
}
```

Using less than in Mongo DB

```
> db.students.find({marks:{$lt:80}})
< {
  _id: ObjectId('6ab21b30555468477e03bd2f'),
  name: 'Rahul',
  age: 22,
  marks: 72,
  department: 'ECE',
  skills: [
    'Python',
    'SQL'
  ]
}
{
  _id: ObjectId('6ab21b30555468477e03bd31'),
  name: 'Neha',
  age: 23,
  marks: 65,
  department: 'ME',
  skills: [
    'AutoCAD',
    'Python'
  ]
}
{
  _id: ObjectId('6ab21b30555468477e03bd32'),
  name: 'Rohit',
  age: 21,
  marks: 78,
  department: 'CSE',
  skills: [
    'Python',
    'AI'
  ]
}
```

Using equal to in Mongo DB

```
> db.students.find({marks:{$eq:78}})
< {
  _id: ObjectId('6ab21b30555468477e03bd32'),
  name: 'Rohit',
  age: 21,
  marks: 78,
  department: 'CSE',
  skills: [
    'Python',
    'AI'
  ]
}
```

Using not equal to in Mongo DB

```
> db.students.find({marks:{$ne:85}})
< {
  _id: ObjectId('6ab21b30555468477e03bd2f'),
  name: 'Rahul',
  age: 22,
  marks: 72,
  department: 'ECE',
  skills: [
    'Python',
    'SQL'
  ]
}
{
  _id: ObjectId('6ab21b30555468477e03bd30'),
  name: 'Priya',
  age: 20,
  marks: 91,
  department: 'CSE',
  skills: [
    'Java',
    'MongoDB',
    'AI'
  ]
}
```


Using greater and equal to in Mongo DB

```
> db.students.find({marks:{$gte:85}})
< {
  _id: ObjectId('6ab21b30555468477e03bd2e'),
  name: 'Amit',
  age: 21,
  marks: 85,
  department: 'CSE',
  skills: [
    'Python',
    'MongoDB',
    'Java'
  ]
}
{
  _id: ObjectId('6ab21b30555468477e03bd30'),
  name: 'Priya',
  age: 20,
  marks: 91,
  department: 'CSE',
  skills: [
    'Java',
    'MongoDB',
    'AI'
  ]
}
```

Using Conditional Operation in Mongo DB

AND Operation:-

```
> db.students.find({$and:[{marks:{$gt:70}},{age:{$lt:23}}]})
< {
  _id: ObjectId('6ab21b30555468477e03bd2e'),
  name: 'Amit',
  age: 21,
  marks: 85,
  department: 'CSE',
  skills: [
    'Python',
    'MongoDB',
    'Java'
  ]
}
{
  _id: ObjectId('6ab21b30555468477e03bd2f'),
  name: 'Rahul',
  age: 22,
  marks: 72,
  department: 'ECE',
  skills: [
    'Python',
    'SQL'
  ]
}
```

Using Conditional Operation in Mongo DB

OR Operation:-

```
> db.students.find({'$or':[{department:"CSE"},{department:"ECA"}]})
< {
  _id: ObjectId('6ab21b30555468477e03bd2e'),
  name: 'Amit',
  age: 21,
  marks: 85,
  department: 'CSE',
  skills: [
    'Python',
    'MongoDB',
    'Java'
  ]
}
{
  _id: ObjectId('6ab21b30555468477e03bd30'),
  name: 'Priya',
  age: 20,
  marks: 91,
  department: 'CSE',
  skills: [
    'Java',
    'MongoDB',
    'AI'
  ]
}
```